

Topic quiz: Projectile motion

Biljana Taneska · Physics teacher

btaneska@vis.ac.at

IBDP Physics SL · Grade 11 · A.1 Kinematics · 10 multiple-choice questions

Name: _____

Class: _____

Date: _____

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use $g = 9.81 \text{ m s}^{-2}$ and show any working next to the question.

Check yourself online: the same quiz with instant feedback and explanations —
<https://physics.taneski.com/quiz-projectile-motion/>



1	2	3	4	5	6	7	8	9	10

Question 1

A ball is dropped from a bench. At the same moment, an identical ball rolls horizontally off the same bench. Air resistance is negligible. Which ball reaches the floor first?

- ☐ (A) Both reach the floor at the same time
- ☐ (B) The dropped ball
- ☐ (C) The ball that rolls off
- ☐ (D) It depends on how fast the second ball rolls

Question 2

A stone is dropped from rest and falls freely for 2.00 s. How far does it fall?

- ☐ (A) 9.81 m
- ☐ (B) 19.6 m
- ☐ (C) 39.2 m
- ☐ (D) 4.91 m

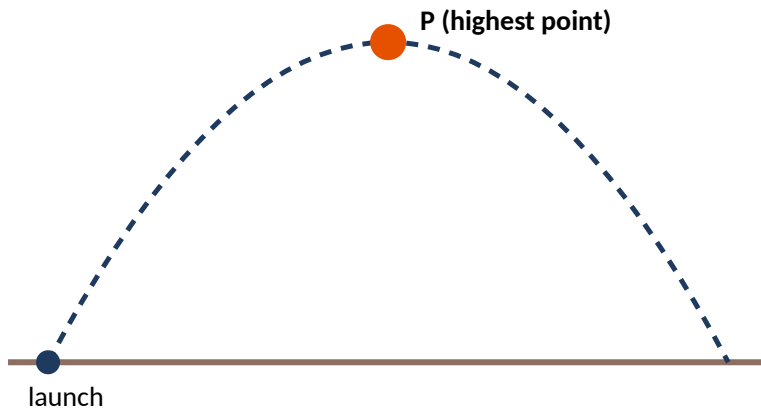
Question 3

A ball rolls off a horizontal bench 0.80 m high with a speed of 2.5 m s^{-1} . How far from the foot of the bench does it land?

- ☐ (A) 0.64 m
- ☐ (B) 0.71 m
- ☐ (C) 1.0 m
- ☐ (D) 2.0 m

Question 4

A ball is launched at an angle above level ground. Air resistance is negligible. Which statement about the ball at its highest point P is correct?



- (A) Its velocity is zero and its acceleration is zero
- (B) Its velocity is zero and its acceleration is 9.81 m s^{-2} downwards
- (C) Its velocity is horizontal and its acceleration is zero
- (D) Its velocity is horizontal and its acceleration is 9.81 m s^{-2} downwards

Question 5

A ball is kicked from level ground at 20.0 m s^{-1} at 30° above the horizontal. What is its maximum height?

- | | |
|------------|------------|
| (A) 5.10 m | (B) 10.2 m |
| (C) 15.3 m | (D) 20.4 m |

Question 6

At a certain speed, a ball launched from level ground at 30° lands 35 m away. At which other launch angle does the same speed give the same range?

- | | |
|----------------|----------------|
| (A) 15° | (B) 45° |
| (C) 60° | (D) 75° |

